

# GROWTH HORMONE

## SOURCE OF SECRETION

- GH is secreted from somatotropes [Acidophilic cell] of Anterior pituitary.

## CHEMISTRY

- GH is protein in nature.
  - Have single-chain polypeptide [191 A.A.]
  - Mw – 21,500

## BLOOD LEVEL & DAILY OUTPUT

- Blood level :—
  - Adult :— 300 ng/dL
  - Children :— 500 ng/dL
- Daily output in Adults – 0.5 – 1.0 mg

## TRANSPORT

- GH is transported in blood by – GH-binding protein (GHBP).

## HALF LIFE

- About 20 minutes.

## METABOLISM

- Degraded in liver and kidney.

## ACTIONS OF GH

- GH is responsible for the general growth of almost all tissue of the body.
- GH ↑es the **SIZE** and **NUMBERS** of the cells by mitotic division.
- GH causes specific differentiation of specific cells:
  - Bone

- Muscles

# 1. ON METABOLISM

**GH ↑es the synthesis of proteins, mobilization of lipids, and conservation of carbohydrates.**

## A. ON PROTEIN METABOLISM

- **GH accelerates the synthesis of proteins by :—**
  - i. ↑es A.A. transports through cell membrane
  - ii. ↑es RNA translation
  - iii. ↑es transcription of DNA to RNA
  - iv. ↓es catabolism of proteins
  - v. Promoting anabolism of proteins

## B. ON FAT METABOLISM

- **GH mobilizes fats from adipose tissue**
  - ↓ *Leads to*
  - **↑es conc. of F.A. in the body fluids**
    - ↷ *Use for production of energy [ $\beta$ -oxidation]*

**NOTE :—** Due to fats mobilization:

- ↷ *Causes production of acetoacetic acid [ketone body] → Leads to ketosis.*
- Also causes **fatty liver**.

## C. ON CARBOHYDRATE METABOLISM

**Major action of GH on carbohydrate is the conservation of glucose.**

- **Effects :—**
  - i. ↓es the peripheral utilization of glucose for energy production  
*[because Acetyl-CoA inhibits glycolytic pathway]*
  - ii. ↑es the deposition of glycogen in the cells.
  - iii. ↓es the uptake of glucose by the cells.
  - iv. **Diabetogenic effect of GH :**
    - Hypersecretion of GH → ↑es blood glucose level
      - ↓ *causes*

- ↑es insulin secretion from  $\beta$ -cell (*GH also stimulates directly*)
- ↓ *due to excess secretion / activity*
- $\beta$ -cells are burnt out at one stage
- ↓ *causes*
- Deficiency of insulin
- ↓ *causes*
- **True diabetes mellitus / Full-blown diabetes mellitus**

## 2. ON BONE

- **GH ACTION ON BONE:**

- **EMBRYONIC STAGE :—** Differentiation + development of bone
- **LATER STAGE :—** Growth of skeleton
  - $\hookrightarrow$  ↑es Length + Thickness

- **IN BONE GH ↑es:**

- i. ↑es synthesis & deposition of protein by chondrocytes & osteogenic cells
- ii. ↑es multiplication of chondrocytes & osteogenic cells
- iii. ↑es new bone formation

by converting chondrocytes  $\longrightarrow$  osteogenic cells

↑es the length of bone until epiphysis fuse with shaft [puberty].

**GH also stimulates osteoblasts strongly:**

- $\hookrightarrow$  Bone continues to grow in thickness throughout life
- $\hookrightarrow$  Membranous bones [Jaw, Skull bone]

## MODE OF ACTION OF GH - SOMATOMEDIN

- **GH acts on bone, growth & protein metabolism through Somatomedin secreted by Liver:**
  - $\hookrightarrow$  Polypeptide ( $M_w - 7,500$ )
  - $\hookrightarrow$  **Two Types:**
    - **Insulin-like growth factor-1 (IGF-I) / [Somatomedin C]**
      - $\hookrightarrow$  *Role:* Bone & Protein metabolism
    - **IGF-II**
      - $\hookrightarrow$  *Role:* Growth of fetus
- **Duration:** The action of Somatomedin C lasts for about 20 hours.

- **Mode of Action of Somatomedin C:**

↪ Acts through 2<sup>nd</sup> messenger called **cAMP**.

## **GROWTH HORMONE RECEPTORS:**

- **Growth Hormone Secretagogue Receptors (GHS):**
  - Transmembrane receptors
  - Belongs to cytokine receptor family
  - Receptors mainly in liver cells.

## **REGULATION OF GH SECRETION**

Hypothalamus and feedback control play important role in regulation of GH secretion.

- **GH secretion stimulated by :—**
  - i. Hypoglycemia
  - ii. Fasting
  - iii. Starvation
  - iv. Exercise
  - v. Stress and Trauma
  - vi. Initial stage of sleep
- **GH secretion inhibited by :—**
  - i. Hyperglycemia
  - ii. ↑es free F.A. in blood
  - iii. Later stage of sleep

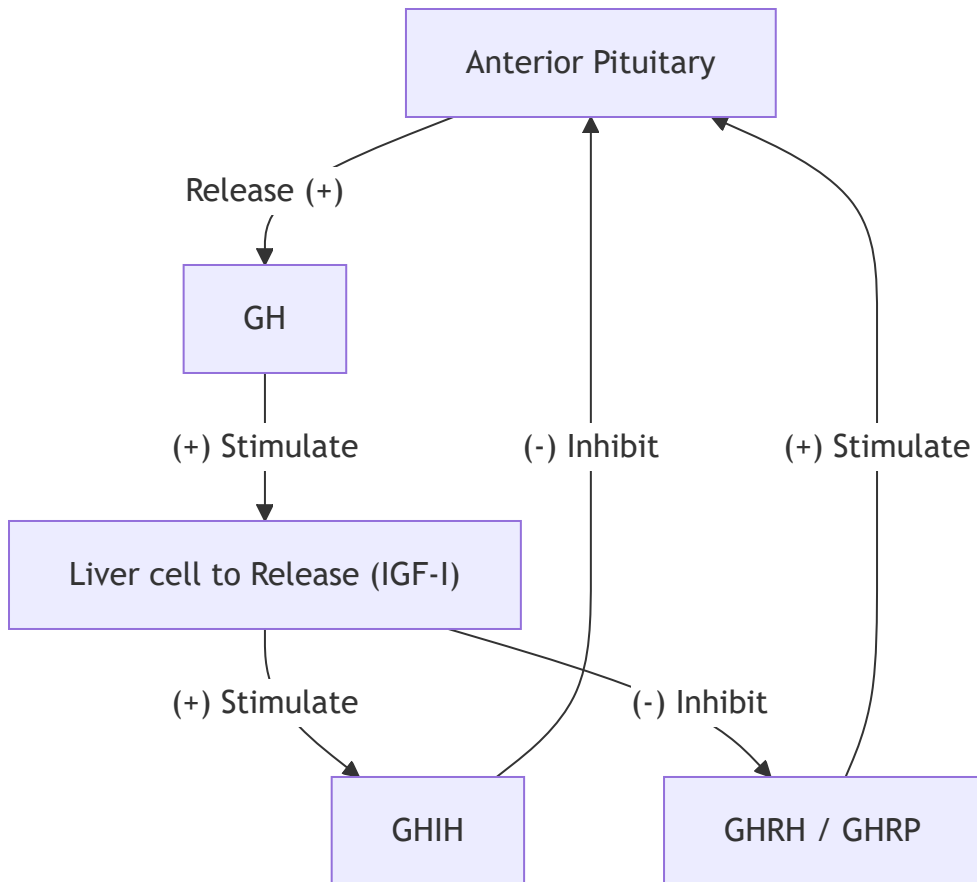
## **REGULATION OF GH SECRETION**

### **1. HYPOTHALAMUS (*by 3 hormones*)**

- **Increase GH secretion:**
  - **GHRH** (*GH-Releasing Hormone*)
  - **GHRP** (*GH-Releasing Polypeptide*)
- **Decrease GH secretion:**
  - **GHIH** (*GH-Inhibitory Hormone*)

**GHRELIN :—** Promotes secretion of GH.

## 2. FEEDBACK CONTROL (*GH-secretion under -ve feedback control*)



## ACTIVITY OF ANTERIOR PITUITARY

| HYPER-ACTIVITY (↑es GH)  | HYPO-ACTIVITY (↓es GH) |
|--------------------------|------------------------|
| 1. Gigantism             | 1. Pituitary Dwarfism  |
| 2. Acromegaly            |                        |
| 3. Acromegalic Gigantism |                        |